

GitHub 101 cheat Sheet

Git is the free and open-source distributed version control system that's responsible for everything GitHub related that happens locally on your computer. This cheat sheet features the most important and commonly used Git commands for easy reference.

INSTALLATION & GUIs

GitHub for Windows <https://windows.github.com>

Git for All Platforms <http://git-scm.com>

SETUP

Configuring user information used across all local repositories

git config --global user.name "[first name last name]"

set a name that is identifiable for credit when review version history

git config --global user.email "[valid-email]"

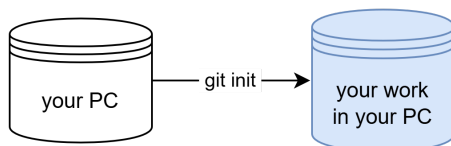
set an email address that will be associated with each history marker

SETUP & INIT

Configuring user information, initializing and cloning repositories

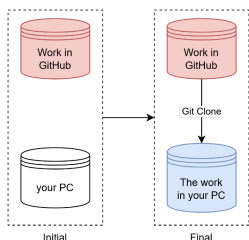
git init

initialize an existing directory as a Git repository



git clone [url]

retrieve an entire repository from a hosted location via URL



STAGE & SNAPSHOT

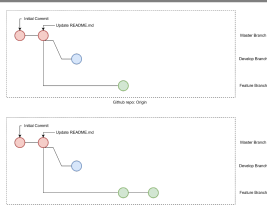
Working with snapshots and the Git staging area

git add [file]

add a file as it looks now to your next commit (stage)

git commit -m "[descriptive message]"

commit your staged content as a new commit snapshot



BRANCH & MERGE

Isolating work in branches, changing context, and integrating changes

git branch [branch-name]

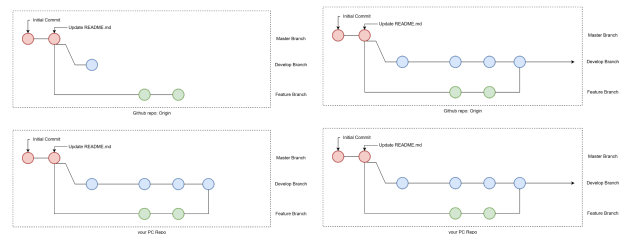
create a new branch at the current commit

git checkout

switch to another branch and check it out into your working directory

git merge [branch]

merge the specified branch's history into the current one



SHARE & UPDATE

Retrieving updates from another repository and updating local repos

git fetch [alias]

fetch down all the branches from that Git remote

git merge [alias]/[branch]

merge a remote branch into your current branch to bring it up to date

git push [alias] [branch]

Transmit local branch commits to the remote repository branch

git pull

fetch and merge any commits from the tracking remote branch

git rebase [branch]

apply any commits of current branch ahead of specified one

git reset --hard [commit]

clear staging area, rewrite working tree from specified commit

